

Name: A. Abhishek Reddy

Rg no:192424192

Activity-1

R-Programming

Scenario 1: Student Academic Performance Dashboard

A. Code

```
Attendance <- c(90, 85, 75, 95)
```

```
Marks <- c(88, 80, 70, 92)
```

```
Department <- c("CSE", "AI", "IT", "ECE")
```

```
col <- c("red", "blue", "green", "orange")
```

```
plot(Attendance, Marks,
```

```
  col = col,
```

```
  pch = 19,
```

```
  main = "Student Academic Performance Dashboard",
```

```
  xlab = "Attendance (%)",
```

```
  ylab = "Internal Marks")
```

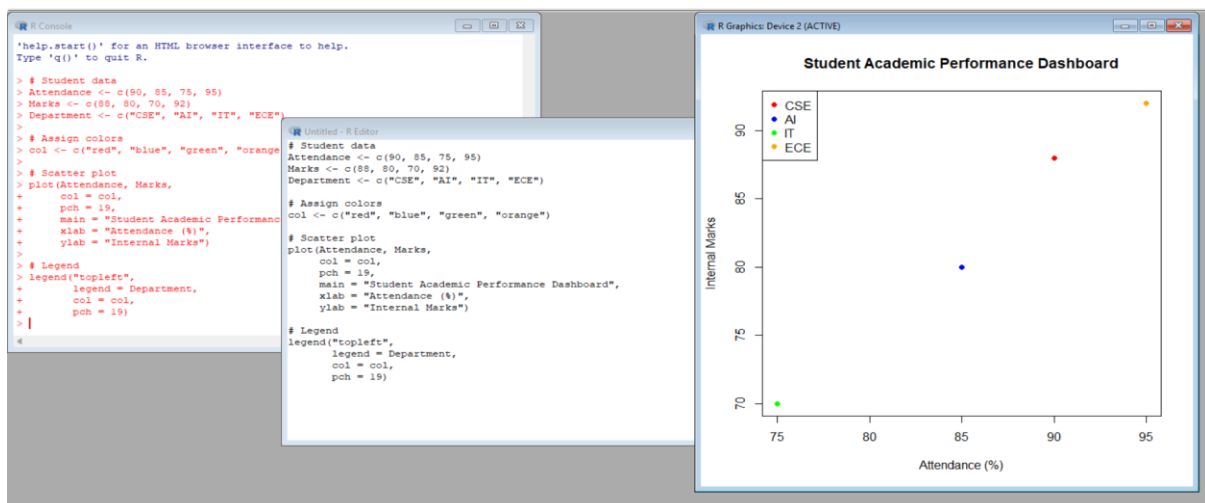
```
legend("topleft",
```

```
  legend = Department,
```

```
  col = col,
```

```
  pch = 19)
```

Output:



Scenario 2: Electric Vehicle Market Analysis

A. Code

```
Battery <- c(30, 40, 50, 60)
Price <- c(10, 15, 20, 25)
Range <- c(200, 300, 400, 500)
Manufacturer <- c("A", "B", "C", "D")
col <- c("red", "blue", "green", "orange")
symbols(Battery, Price,
        circles = Range/50,
        bg = col,
        inches = 0.3,
        xlab = "Battery Capacity (kWh)",
        ylab = "Price",
        main = "Electric Vehicle Market Analysis")
legend("topleft",
        legend = Manufacturer,
        fill = col)
```

Output:



Scenario 3: Hospital Patient Admission Analysis

A. Code

```
Department <- c("Cardiology", "Neurology", "Orthopedics", "Pediatrics")
```

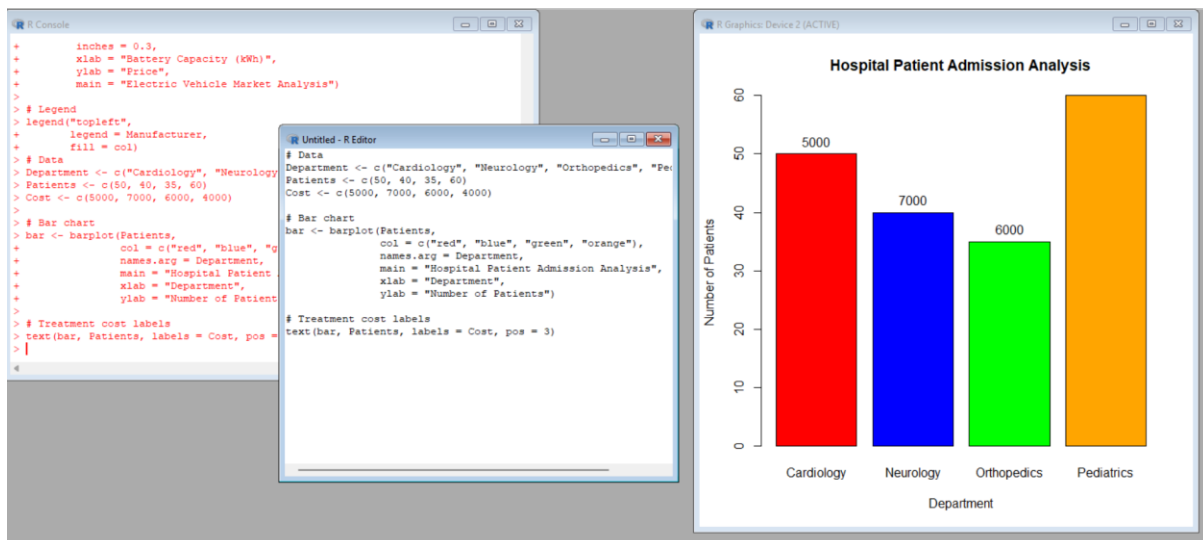
```
Patients <- c(50, 40, 35, 60)
```

```
Cost <- c(5000, 7000, 6000, 4000)
```

```
bar <- barplot(Patients,  
               col = c("red", "blue", "green", "orange"),  
               names.arg = Department,  
               main = "Hospital Patient Admission Analysis",  
               xlab = "Department",  
               ylab = "Number of Patients")
```

```
text(bar, Patients, labels = Cost, pos = 3)
```

Output:



Scenario 4: Weather Monitoring System

A. Code

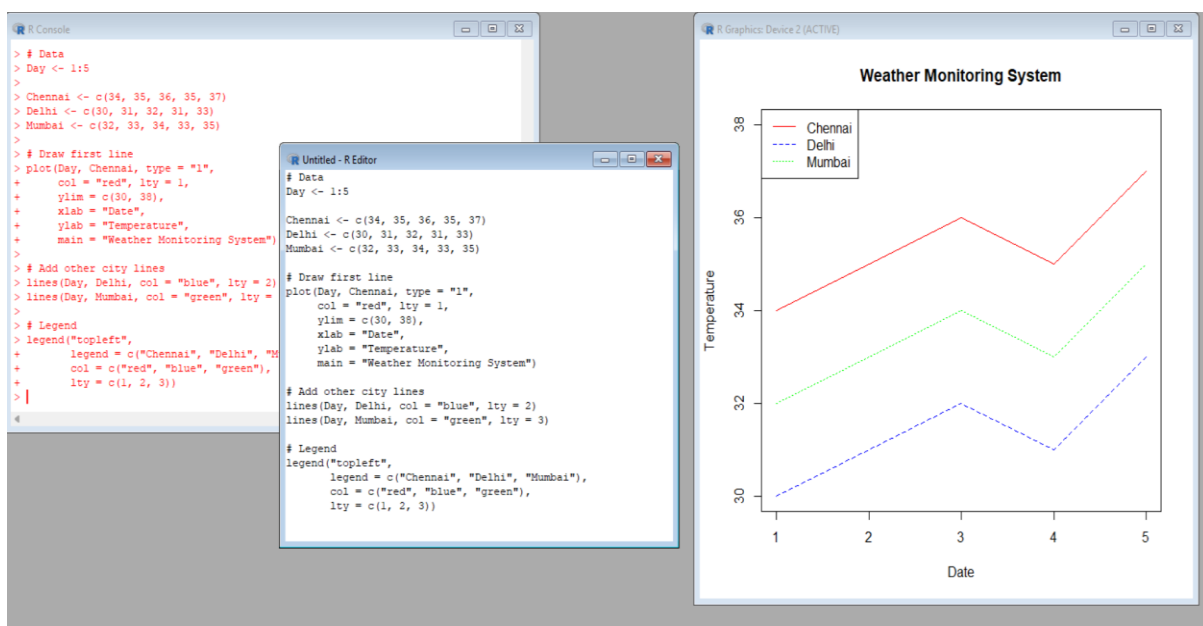
```
Day <- 1:5
Chennai <- c(34, 35, 36, 35, 37)
Delhi <- c(30, 31, 32, 31, 33)
Mumbai <- c(32, 33, 34, 33, 35)

plot(Day, Chennai, type = "l",
     col = "red", lty = 1,
     ylim = c(30, 38),
     xlab = "Date",
     ylab = "Temperature",
     main = "Weather Monitoring System")

lines(Day, Delhi, col = "blue", lty = 2)
lines(Day, Mumbai, col = "green", lty = 3)

legend("topleft",
     legend = c("Chennai", "Delhi", "Mumbai"),
     col = c("red", "blue", "green"),
     lty = c(1, 2, 3))
```

Output:



Scenario 5: Supermarket Sales Analysis

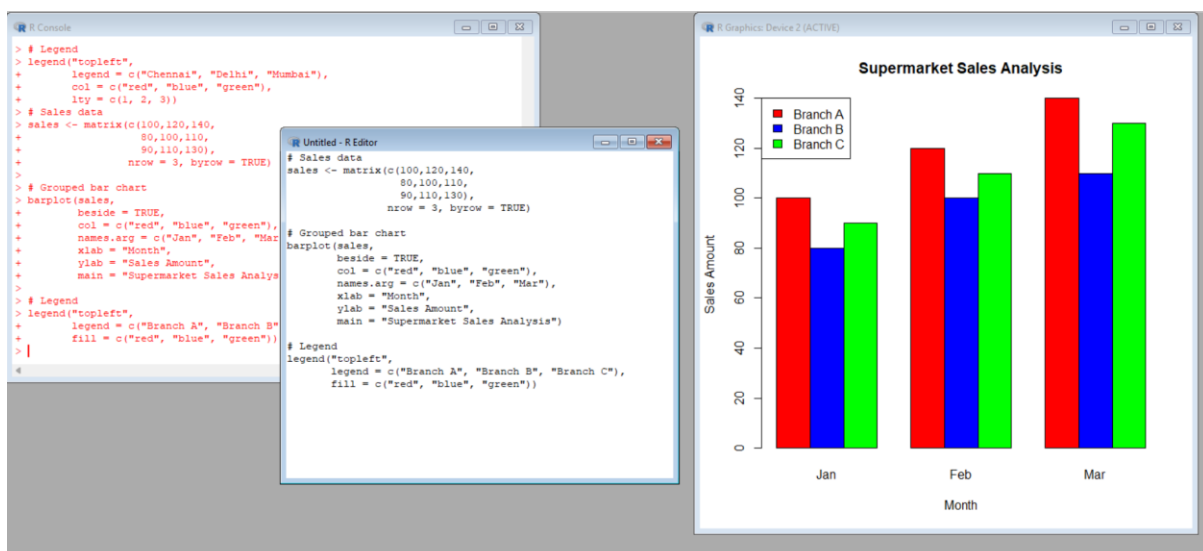
A. Code

```
sales <- matrix(c(100,120,140,
                 80,100,110,
                 90,110,130),
               nrow = 3, byrow = TRUE)

barplot(sales,
        beside = TRUE,
        col = c("red", "blue", "green"),
        names.arg = c("Jan", "Feb", "Mar"),
        xlab = "Month",
        ylab = "Sales Amount",
        main = "Supermarket Sales Analysis")

legend("topleft",
       legend = c("Branch A", "Branch B", "Branch C"),
       fill = c("red", "blue", "green"))
```

Output:



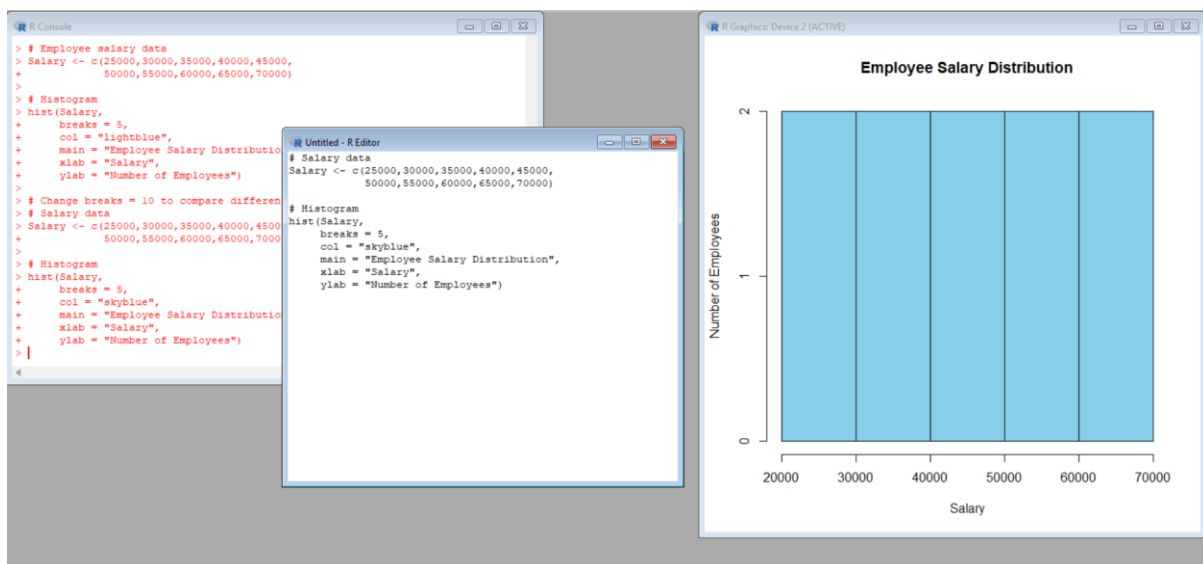
Scenario 6: Employee Salary Distribution

A. Code

```
Salary <- c(25000,30000,35000,40000,45000,  
           50000,55000,60000,65000,70000)
```

```
hist(Salary,  
     breaks = 5,  
     col = "skyblue",  
     main = "Employee Salary Distribution",  
     xlab = "Salary",  
     ylab = "Number of Employees")
```

Output:



Scenario 7: Smartphone Market Comparison

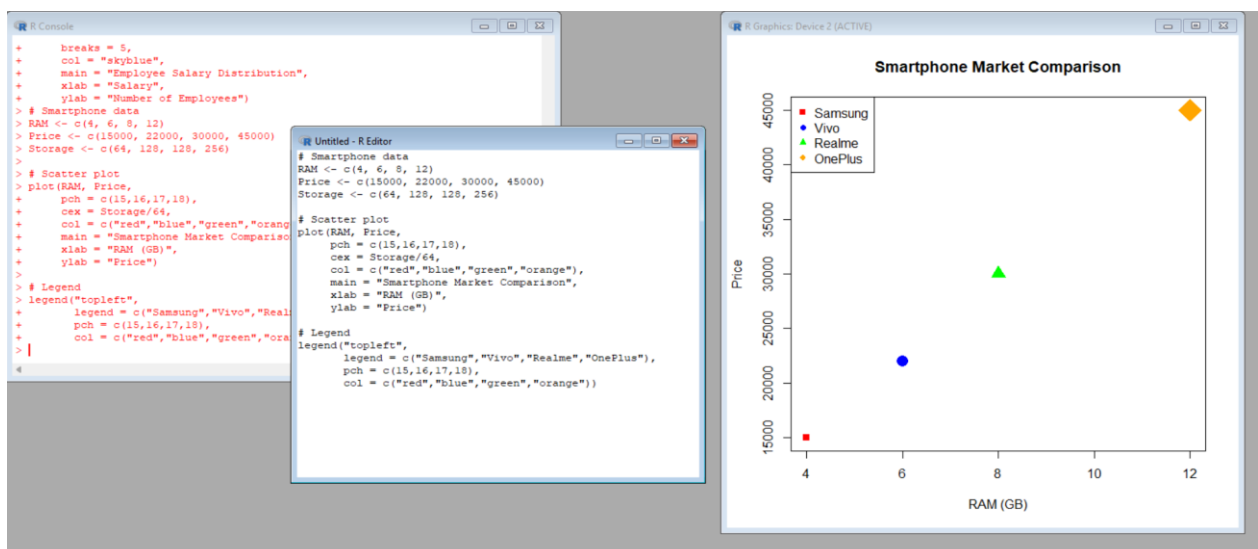
A. Code

```
RAM <- c(4, 6, 8, 12)
Price <- c(15000, 22000, 30000, 45000)
Storage <- c(64, 128, 128, 256)

plot(RAM, Price,
     pch = c(15,16,17,18),
     cex = Storage/64,
     col = c("red","blue","green","orange"),
     main = "Smartphone Market Comparison",
     xlab = "RAM (GB)",
     ylab = "Price")

legend("topleft",
     legend = c("Samsung","Vivo","Realme","OnePlus"),
     pch = c(15,16,17,18),
     col = c("red","blue","green","orange"))
```

Output:



Scenario 8: Online Movie Rating Analysis

A. Code

```
rating <- matrix(c(4.5, 3.8, 4.2, 4.8), nrow = 2)
```

```
heatmap(rating,
```

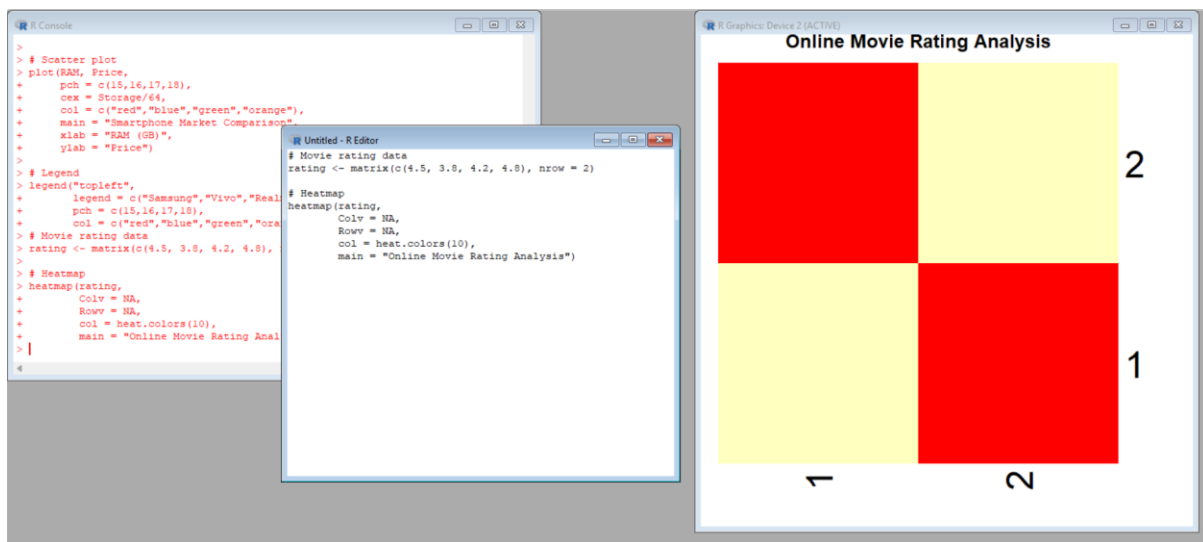
```
  Colv = NA,
```

```
  Rowv = NA,
```

```
  col = heat.colors(10),
```

```
  main = "Online Movie Rating Analysis")
```

Output:

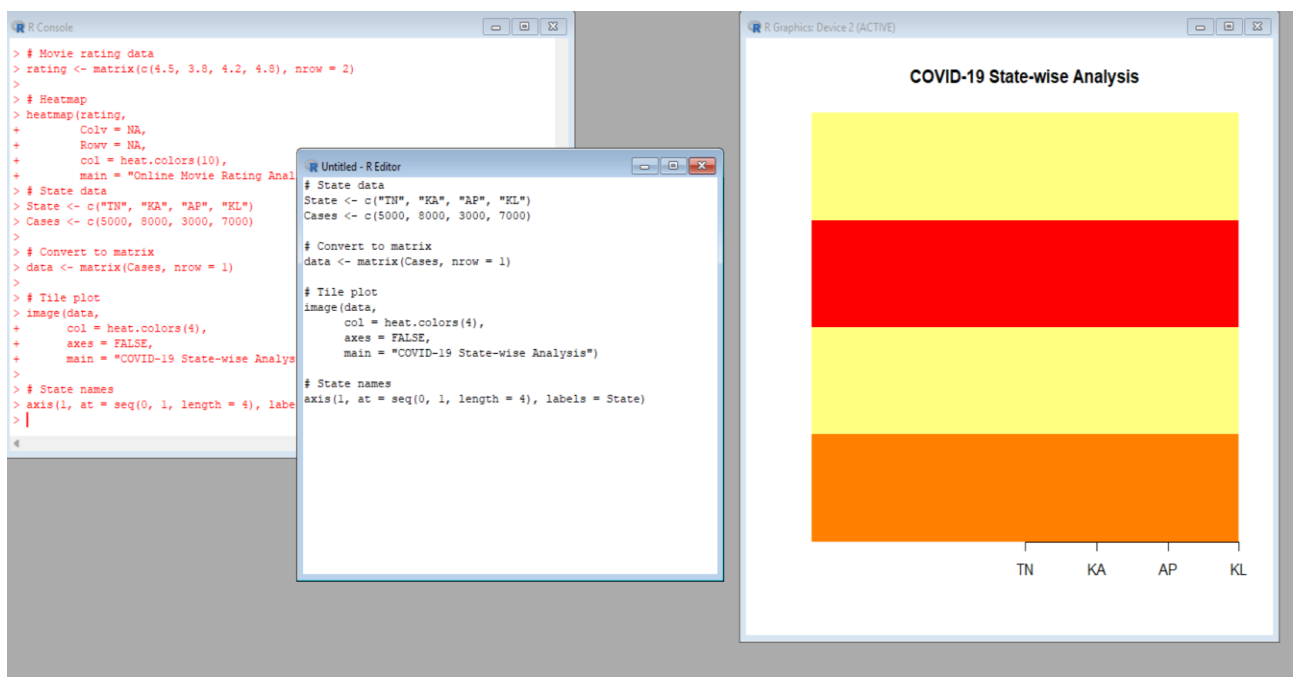


Scenario 9: COVID-19 State-wise Analysis

A. Code

```
State <- c("TN", "KA", "AP", "KL")
Cases <- c(5000, 8000, 3000, 7000)
data <- matrix(Cases, nrow = 1)
image(data,
      col = heat.colors(4),
      axes = FALSE,
      main = "COVID-19 State-wise Analysis")
axis(1, at = seq(0, 1, length = 4), labels = State)
```

Output:



Scenario 10: Iris Flower Classification

A. Code

```
plot(iris$Sepal.Length,  
     iris$Petal.Length,  
     col = as.numeric(iris$Species),  
     pch = as.numeric(iris$Species),  
     cex = iris$Sepal.Width/2,  
     xlab = "Sepal Length",  
     ylab = "Petal Length",  
     main = "Iris Flower Classification")  
  
legend("topleft",  
      legend = levels(iris$Species),  
      col = 1:3,  
      pch = 1:3)
```

Output:

